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Graded Assignment • 30 min

Due Mar 16, 11:59 PM CET

1. Evolutionary science tries to answer all of the following questions EXCEPT: 1 point

- How and why have the properties of the units of nature changed over time?
- What are the biochemical mechanisms for transport of ions across cell membranes?
- What are the genealogical relationships between the units of nature - species and populations?
- What are the units of nature? That is, what species or assemblages exist in nature?

2. A phylogenetic tree is: 1 point

- a hypothesis about the evolutionary relationships of taxa
- a diagram depicting the network of natural affinities based on similarities
- a description of a species
- an explanation of the process of speciation

3. Darwin's theories as set forth in the *Origin of Species* included all EXCEPT which one of the following? 1 point

- discussion of new species discovered on his voyage and the role of natural selection in their origin
- discussion of a struggle for existence that would account for fossils and extinct organisms
- ideas on how life originated on earth from non-living elements and ingredients
- observations that supported his idea of the descent of life with modifications through time

4. In cladograms used to build the Tree of Life: 1 point

- Descendants from a common ancestor that share derived characters are more distantly related to each other than to descendants that have the primitive condition of that character.
- Primitive characters are very informative about relationships within a group such as the vertebrates because every group within the vertebrate tree shares this feature.
- There is no way to include extinct species in a cladogram.
- Taxa are grouped on the basis of shared *derived* similarities, not shared *primitive* similarities.

5. Which of the following is NOT a reason that "model organisms" are valued by evolutionary biologists? 1 point

- They are comparatively easy to work with and often have short generation times.
- Their unique structure and biochemistry allows evolutionary biologists to segregate them and document them as extinction points on the "tree of life".
- They can give data on evolutionary pathways that delineate the branches on the tree of life.
- Their biochemical systems can be compared to human systems.

6. Constructing the most plausible cladogram: 1 point

- involves examining alternative trees and selecting the one that shows the greatest number of branch points as the most likely evolutionary pattern
- involves the use of parsimony to select the most simple arrangement of characters within a branch
- involves the use of parsimony to select the most complex arrangement of characters within a branch
- involves plotting characters on cladograms to show only convergence points and then selecting the cladogram that displays the greatest number of transformations as the most likely evolutionary pattern

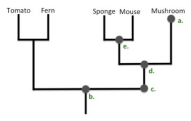
7. Which of the following is NOT an accepted mechanism for speciation? 1 point

- sexual selection
- maximum parsimony
- natural selection
- genetic drift

8. Which of the following is not a pattern that Darwin uncovered while on his voyage on the Beagle? 1 point

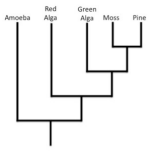
- Extinct organisms, such as the glyptodont and giant sloths, are similar to organisms that are alive today.
- Organisms can be placed in "natural groups" for classification purposes.
- Similar yet distinct organisms, such as the Galapogos tortoises, occupy different islands.
- Similar yet distinct organisms, such as different types of rheas, occupy different geographic areas.

9. Which of the five marks on the tree below corresponds to the most recent common ancestor of a mushroom and a sponge? 1 point



- a
- b
- c
- d
- e

10. By reference to the tree below, which of the following is an accurate statement of relationships? 1 point



- A green alga is more closely related to a red alga than to a moss.
- A green alga is more closely related to a moss than to a red alga.
- A green alga is equally related to a red alga and a moss.
- A green alga is related to a red alga, but is not related to a moss.